EXPERIMENT 6

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D15A - 44

AIM: Apply K-Means and Hierarchical Clustering on sample datasets

THEORY:

1. Dataset Source

Dataset: CEEW India Residential Energy Survey Microdata

Source: Council on Energy, Environment and Water

Source link:
https://www.kaggle.com/code/raitest/india-residential-energy-survey-ires-2020-eda/input?select=CEEW+-+IRES+Data.csv

2. Dataset Description

For this experiment, the following numeric variables were selected to capture energy consumption behaviour:

avg_monthly_bill

q302_grid_hrs_no

q308_grid_voltage_low_app

q405_c_led_bulb_no

q410_ceiling_fan_no

q467_fridge_no

q460_tv_no

q232_mobile_smart_n

These variables represent:

- Financial electricity burden
- Reliability of supply
- Appliance intensity
- Digital penetration

After removing missing values, clustering was performed on approximately 2987 households.

3. Mathematical Formulation of the Algorithms

K-Means Clustering

Objective:

Minimize within-cluster sum of squares:

Minimize

$$\Sigma \Sigma \|x_i - \mu_k\|^2$$

Where:

x_i is a data point

μ_k is the centroid of cluster k

Steps:

1. Initialize k centroids
2. Assign each point to nearest centroid
3. Update centroids
4. Repeat until convergence
5. Distance metric used: Euclidean distance
6. Hierarchical Clustering
7. Agglomerative approach used:
8. Start with each point as its own cluster
9. Iteratively merge closest clusters
10. Ward linkage method minimizes variance within clusters

Ward linkage minimizes:

Increase in total within-cluster variance after merging clusters

4. Algorithm Limitations

- K-Means assumes spherical clusters

- Sensitive to feature scaling
- Sensitive to extreme outliers
- Requires pre-specifying number of clusters
- Hierarchical clustering computationally expensive for very large datasets

5. Methodology / Workflow

1. Import required libraries
2. Load dataset
3. Select energy consumption and appliance variables
4. Remove missing values
5. Apply StandardScaler
6. Determine optimal number of clusters using Elbow Method
7. Evaluate cluster quality using Silhouette Score
8. Train final K-Means model
9. Assign cluster labels
10. Visualize clusters using PCA projection
11. Perform Hierarchical Clustering using Ward linkage
12. Plot dendrogram
13. Interpret cluster profiles

6. Performance Analysis

Optimal number of clusters selected: 3

Cluster distribution:

- Cluster 0: 2200 households
- Cluster 2: 746 households
- Cluster 1: 41 households

Interpretation of cluster sizes:

- Majority cluster represents dominant consumption pattern
- Moderate-sized cluster indicates distinct mid-level segment
- Small cluster represents extreme or high-intensity segment

Cluster Behavioural Segmentation:

Cluster 0

- Likely low to moderate monthly bills
- Fewer appliances
- Limited digital intensity

- Represents majority consumption class

Cluster 2

- Moderate to high electricity usage
- Higher appliance ownership
- Better infrastructure conditions
- Emerging digital adoption

Cluster 1

- Very high consumption intensity
- High appliance penetration
- Higher financial electricity burden
- Likely urban or high-asset households
- Silhouette score indicated meaningful separation between clusters, validating segmentation quality.
- Hierarchical dendrogram supported multi-level grouping structure, confirming existence of distinct consumption tiers.

7. Hyperparameter Selection and Model Decisions

1. Elbow method used to determine optimal k
2. Silhouette score used to validate cluster compactness
3. Ward linkage selected for hierarchical clustering due to variance minimization property
4. StandardScaler applied to ensure equal feature contribution
5. No supervised tuning was required as clustering is unsupervised.

OUTPUT:

1. Three distinct household energy consumption segments identified.
2. Clear structural differentiation in appliance ownership and electricity intensity.
3. Extreme high-consumption micro-segment identified separately.
4. PCA visualization confirmed cluster separation in reduced dimensional space.
5. Hierarchical dendrogram validated multi-level grouping structure.

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.decomposition import PCA
from scipy.cluster.hierarchy import linkage, dendrogram
```

```python
df = pd.read_csv("CEEW - IRES Data.csv", low_memory=False)
print("Dataset shape:", df.shape)
```

```
Dataset shape: (14851, 517)
```

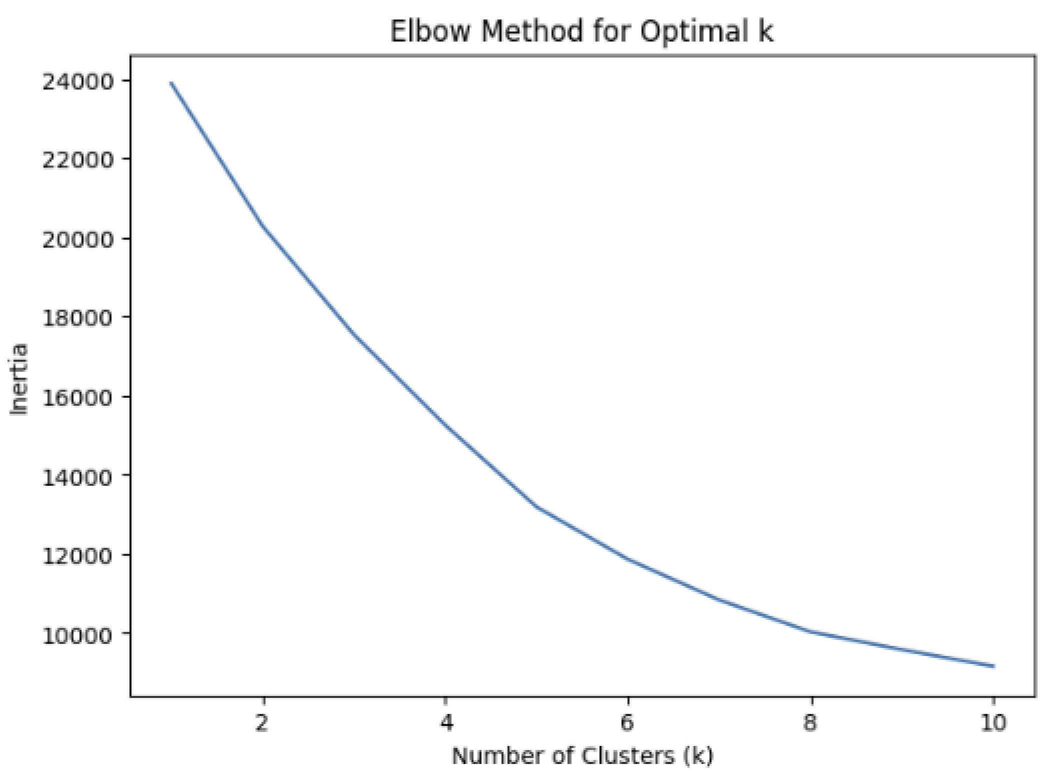
```python
fields = [
    'avg_monthly_bill',
    'q302_grid_hrs_no',
    'q308_grid_voltage_low_app',
    'q405_c_led_bulb_no',
    'q410_ceiling_fan_no',
    'q467_fridge_no',
    'q460_tv_no',
    'q232_mobile_smart_n'
]

df_cluster = df[fields].dropna()

print("Clustering dataset shape:", df_cluster.shape)
df_cluster.describe()
```

```
Clustering dataset shape: (2987, 8)
```

	avg_monthly_bill	q302_grid_hrs_no	q308_grid_voltage_low_app	q405_c_led_bulb_no	q410_ceiling_fan_no	q467_fridge_no	q460_
count	2987.000000	2987.000000	2987.000000	2987.000000	2987.000000	2987.000000	2987.0
mean	1016.676152	22.180449	0.503180	3.155340	2.273853	1.014061	1.0
std	884.715837	3.174419	1.449845	2.281432	1.111347	0.120572	0.2
min	0.000000	7.000000	0.000000	0.000000	1.000000	1.000000	1.0
25%	400.000000	22.000000	0.000000	2.000000	1.000000	1.000000	1.0
50%	796.666690	24.000000	0.000000	3.000000	2.000000	1.000000	1.0

```python
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df_cluster)
```

```python
inertia = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    inertia.append(kmeans.inertia_)

plt.figure(figsize=(7,5))
plt.plot(range(1, 11), inertia)
plt.xlabel("Number of Clusters (k)")
plt.ylabel("Inertia")
plt.title("Elbow Method for Optimal k")
plt.show()
```



```
for k in range(2, 7):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = kmeans.fit_predict(X_scaled)
    score = silhouette_score(X_scaled, labels)
    print(f"k = {k}, Silhouette Score = {score:.4f}")
```

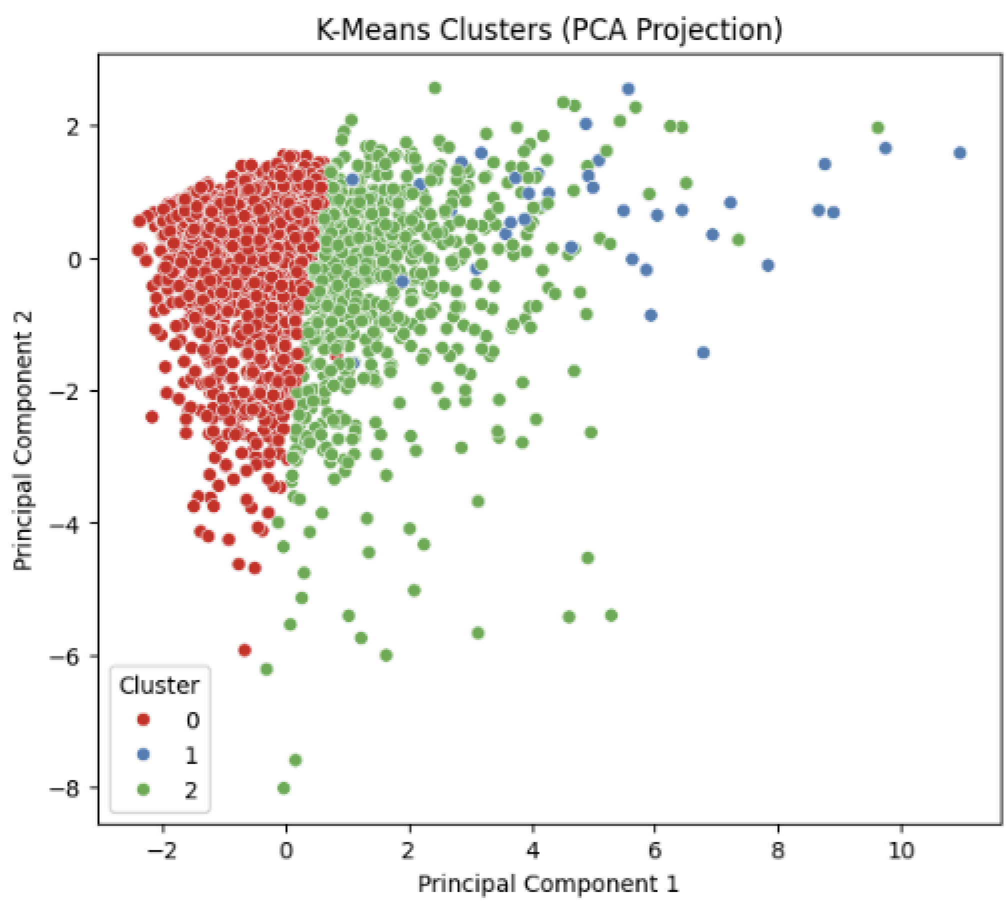
```
k = 2, Silhouette Score = 0.5586
k = 3, Silhouette Score = 0.3035
k = 4, Silhouette Score = 0.3111
k = 5, Silhouette Score = 0.2572
k = 6, Silhouette Score = 0.2679
```

```
optimal_k = 3

kmeans = KMeans(n_clusters=optimal_k, random_state=42, n_init=10)
clusters = kmeans.fit_predict(X_scaled)

df_cluster['Cluster'] = clusters

df_cluster['Cluster'].value_counts()
```

	count
Cluster	
0	2200
2	746
1	41

dtype: int64

```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

plt.figure(figsize=(7,6))
sns.scatterplot(x=X_pca[:,0], y=X_pca[:,1], hue=clusters, palette='Set1')
plt.title("K-Means Clusters (PCA Projection)")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend(title="Cluster")
plt.show()
```



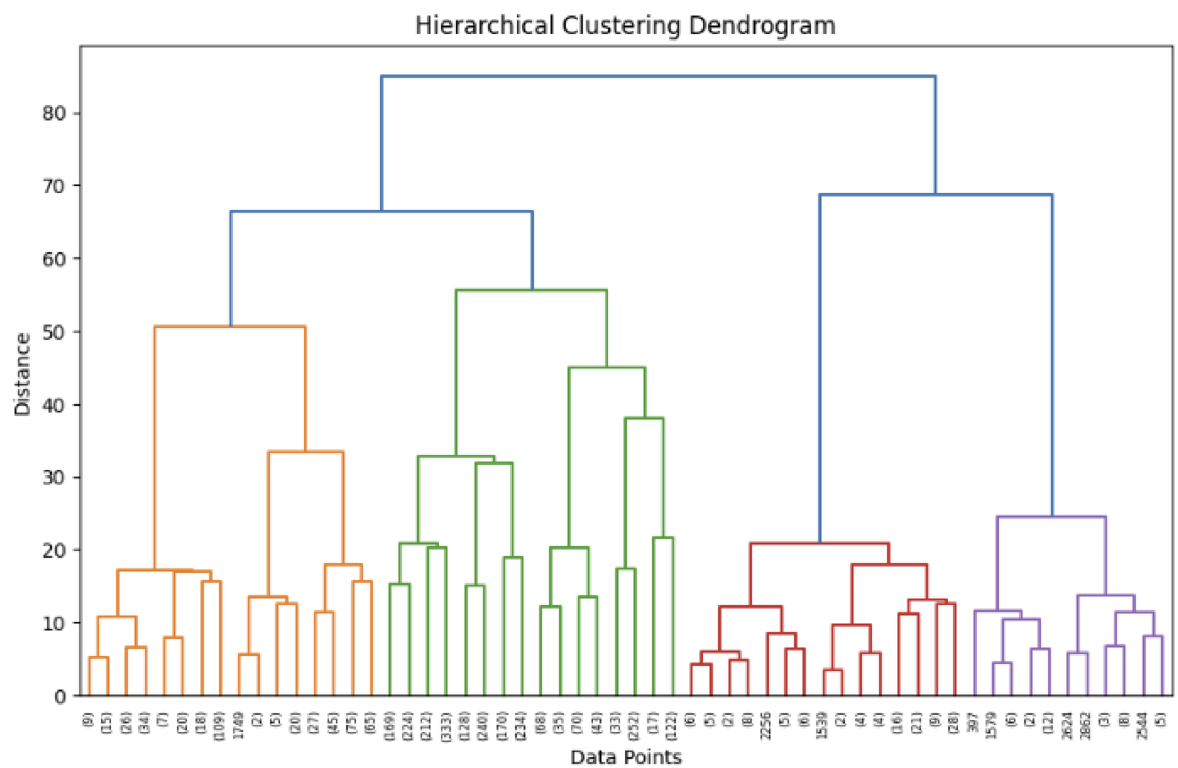

```
cluster_profile = df_cluster.groupby('Cluster').mean()
cluster_profile
```

Cluster	avg_monthly_bill	q302_grid_hrs_no	q308_grid_voltage_low_app	q405_c_led_bulb_no	q410_ceiling_fan_no	q467_fridge_no	q460_tv_no	q232_mobile_smart_n
0	813.885000	22.347273	0.374091	2.476818	1.847727	1.00000	1.002727	1.625909
1	1414.308958	23.000000	0.609756	5.146341	3.682927	2.02439	1.512195	2.853659
2	1592.865951	21.643432	0.878016	5.046917	3.453083	1.00000	1.151475	2.651475

t steps: Generate code with cluster_profile New interactive sheet

```
linked = linkage(X_scaled, method='ward')
```

```
plt.figure(figsize=(10,6))
dendrogram(linked, truncate_mode='level', p=5)
plt.title("Hierarchical Clustering Dendrogram")
plt.xlabel("Data Points")
plt.ylabel("Distance")
plt.show()
```



CONCLUSION:

1. Household energy consumption patterns form clear structural segments.
2. Majority of households fall under low to moderate consumption category.
3. A smaller segment exhibits significantly higher appliance intensity and electricity burden.
4. Clustering reveals stratification in infrastructure reliability and digital penetration.

Energy consumption behaviour is not uniform and can be segmented into identifiable structural tiers.This experiment demonstrates how unsupervised learning can uncover hidden behavioural patterns in large-scale household energy datasets, enabling structured interpretation of consumption intensity and infrastructure-based segmentation.